

University of Dhaka
Affiliated Engineering Colleges
Department of Computer Science and Engineering
1st Year 1st Semester B.Sc. in CSE Final Examination, 2022
MATH-1105: Differential and Integral Calculus

Total Marks: 70

Time: 3 Hours

(Answer any 5 (Five) of the following Questions)

1. a) Describe Linear Approximation with examples. 3
 b) Test the Continuity and Differentiability of the following function at $x = \frac{1}{2}$ 6

$$f(x) = \begin{cases} x & \text{when } 0 \leq x < \frac{1}{2} \\ 1 - x & \text{when } \frac{1}{2} \leq x < 1 \end{cases}$$

 c) Prove that If a function is differentiable at any point then the function is continuous at that point. 5
2. a) The value of a Car is \$21500. It loses 12% of it's value every year. 7
 i) Write a function that represents the value (in Dollar) of the car after t years.
 ii) Find the approximate monthly percent decrease in value.
 b) Determine the maximum and minimum values of $f(x) = x^3 - 9x^2 + 15x - 3$ 4
 c) Sketch the graph of $f(x) = x(x - 2)^{\frac{2}{3}}$ 3
3. a) Find the surface area of the solid generated by revolving the parabola $y^2 = 4ax$ between $x = 0$ and $x = a$ about the X axis. 7
 b) Find the volume of surface of revolution of cycloid $x = a(\theta + \sin \theta)$ and $y = a(1 - \cos \theta)$ rounding it's highest point of tangent. 7
4. a) A function $f(x)$ is defined as follows: 6

$$f(x) = \begin{cases} x & \text{when } 0 < x < 1 \\ 2 - x & \text{when } 1 \leq x \leq 2 \\ x - \frac{x^2}{2} & \text{when } x > 2 \end{cases}$$

 Test the differentiability of $f(x)$ at $x = 1$ and $x = 2$.
 b) Find the linear approximation of $f(x) = \sqrt{x}$ at $x = 9$ with graph. 4
 c) State L' Hospital's Rule. Evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{1}{x^2} \log(1 + x) \right]$ 4
5. a) Evaluate the following integrals (any two): 3.5

$$\text{i) } \int \frac{1}{\sqrt{(x^2 - a^2)}} dx \quad \text{ii) } \int \frac{1}{x^4 + 1} dx \quad \text{iii) } \int \frac{1}{(x+2)\sqrt{(x^2 - 5)}} dx$$
 x 2 = 7
 b) Evaluate the following integrals (any two): 3.5

$$\text{i) } \int \frac{1}{\sqrt{(a-x)(x-b)}} dx \quad \text{ii) } \int_0^1 x^3 \sqrt{1 + 3x^4} dx \quad \text{iii) } \int \frac{1}{3 + 2 \sin x} dx$$
 x 2 = 7
6. a) Define Gamma and Beta function. 2
 b) Prove that $\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$ 6
 c) Show that $\int_0^{\frac{\pi}{2}} \sin^p \theta \cos^q \theta d\theta = \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{2 \Gamma(\frac{p+q+2}{2})}$ 6
7. a) Describe Indeterminate forms with examples. 2
 b) Verify Mean Value theorem of Integrals for $f(x) = x(1 - x)$ on $[0, 1]$ 6
 c) Find the arc length of the cardioid $r = a(1 - \cos \theta)$. 6